

TECHNICAL BULLETIN

DIXIE CHEMICAL

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Propyleneimine (PI) CAS Reg. No. 75-55-8

Typical Properties

Property	Values	Test Method
Appearance/Odor	Water White Liquid, Strong, ammonia-like odor.	D-104 (Visual)
Purity, %	97.5 Min.	D-225
Water, %	1.0 Max.	D-225

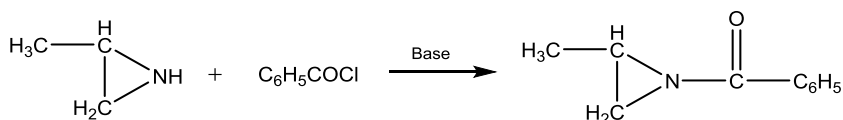
Applications

Propyleneimine is a versatile organic intermediate containing a highly reactive three-membered aziridine ring. Reactions of PI fall into two broad groups: as a secondary amine yielding N-substituted propyleneimines in which the ring remains intact and as a cyclic aziridine involving ring opening reactions including polymerization.

Reactions of PI as a secondary amine

Reaction conditions must be adjusted to avoid simultaneous opening of the aziridine ring. This is accomplished by maintaining minimal reaction temperatures, conducting reactions in alkaline media or by removing acid produced through the use of a suitable acid acceptor such as tertiary amines or caustic. Some examples of this chemistry are shown below:

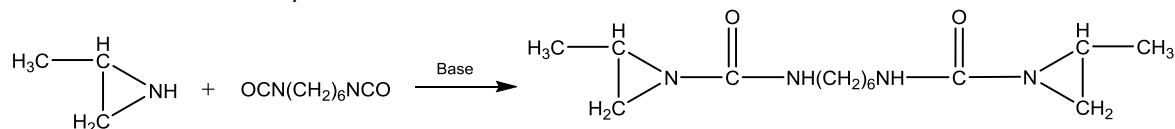
Reaction with acid chlorides:



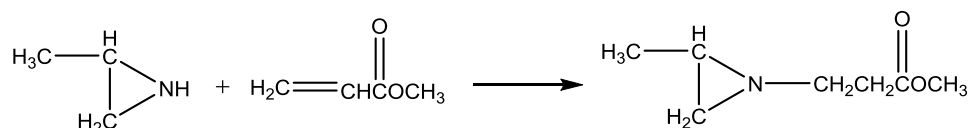
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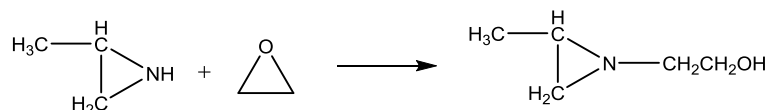
Reaction with isocyanates:



Reaction with α,β -unsaturated compounds (Michael addition)



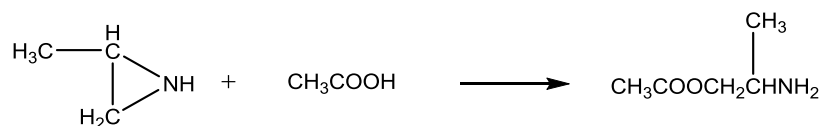
Reaction with epoxides:



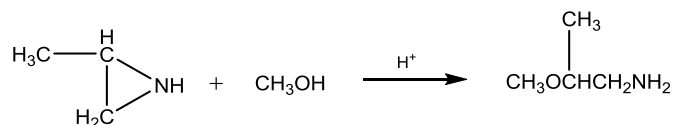
Ring Opening Reactions

Addition products of PI can be obtained via ring opening at the carbon-nitrogen bond by treatment with active hydrogen compounds. Some examples are illustrated below:

Reaction with carboxyl groups:



Reaction of PI with alcohols:

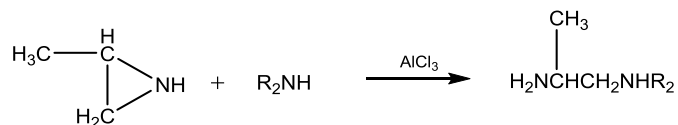


Reaction with amines:

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Propyleneimine has been used extensively as an intermediate for coatings with improved adhesion, water resistance and hardness. PI has also shown wide utility as an intermediate in polymers such as polyamide, polyurethane and polyester to modify the structure or add various functional groups to the polymer. PI is used in adhesives, paper treatment, textile dyeing and printing, petroleum refining, fiber modification, rocket propellant fuel and biological products.

Handling & Storage

Propyleneimine is a potentially hazardous chemical. Read and understand the relevant Safety Data Sheets (SDS) for all products before use. Additional product stewardship information is available upon request.

PI is supplied in specially constructed steel cylinders, which contain a small amount of sodium hydroxide as a stabilizer. PI can be pumped, gravity fed or pulled by vacuum into suitable receiving vessels through flexible 316 stainless steel hoses. PI is a high vapor pressure, flammable material which must be handled with care. Take measures to prevent the buildup of electrostatic charge and use bonding or grounding on vessels. PI will react with acids, including mild acids and acid forming materials. Extensive care must be taken to avoid accidental contact of PI with these materials. PI has a shelf life of 24 months when properly stored in unopened containers.

Health Hazards

Propyleneimine is highly toxic by inhalation, dermal contact and ingestion. PI may cause cancer. Always wear appropriate personal protective equipment as specified in the SDS.

Availability

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Propyleneimine is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie representative for details.